

WASTEWISE SCAN · DESIGN DECISION RECORD

WasteWise Scan shared scan record format (proposal)

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	WSC-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
Source	docs/decisions/0002-scan-record-format.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First proposal of a scan record shared with WasteWise-ml, for joint review

0002: Shared scan record format (proposal)

- **Date:** 2026-09-25
- **Status:** proposed. Proposed, awaiting Amish and a joint review with WasteWise-ml.

■ Context

Requirement R13 asks that low-confidence scans hand off to WasteWise-ml and that scan records use a shared, documented format. WasteWise-ml labels items with the taxonomy in its `ml/data/taxonomy.yaml` (version 0.2, 2026-09-25): a material class, a grade (for plastics, the resin code, for example "PET (1)" or "other or unknown (7)"), and two non-class outputs, "unsure" and "hazard". WasteWise Scan produces a resin, a confidence, a local price grade and an eight-band spectrum. ReflowEconomy will later want the resin of what it buys.

This note is written in the WasteWise Scan repo only. The WasteWise-ml repo was read and not changed. Nothing here is agreed until Amish and the WasteWise-ml side accept it.

■ Options considered

1. **One shared record for both tools (recommended).** Both write the same fields; each leaves the fields it cannot fill empty. Combining a photo and a scan of the same item is a join on `pair_id`.
2. **Separate records with a mapping table.** Each tool keeps its own format and a converter joins them. Less coordination now, more drift later.
3. **Adopt an external schema.** No open schema for field resin scans was found in this session; this can be revisited.

■ Proposed record

Recommendation: option 1, with the fields in Table 1. Exports are CSV (one row per record, spectrum as eight columns) and JSON Lines (one object per line). Labels reuse the WasteWise-ml taxonomy strings exactly, so no mapping is needed.

Table 1. Shared scan record, version 0.1 (proposed).

Field	Type	Filled by	Meaning
record_version	text	both	"0.1"
source	text	both	"wws" (WasteWise Scan) or "wml" (WasteWise-ml)
device_id	text	both	Short random ID set at first start; no personal data
record_id	text	both	device_id plus a sequence number, unique per device
time_utc	text	both	ISO 8601, UTC, to the second
site_id	text	both, optional	The price table in use (cooperative or buyer site); set by the partner
pair_id	text	both, optional	Same value on a photo and a scan of the same item, entered or scanned by the user; empty if unpaired
material_class	text	both	Taxonomy class; WasteWise Scan writes "plastic" or leaves it empty for "unknown"
grade	text	both	Taxonomy grade, for example "PET (1)", "PP (5)", "other or unknown (7)"
output	text	both	"answer", "unsure" or "hazard"; WasteWise Scan never writes "hazard"
confidence	number	both	0 to 1
price_grade	text	both, optional	Local grade from the site price table, for example "A"
model_version	text	both	Classifier or model version
wws_bands_nm	list of 8 numbers	WasteWise Scan	850, 940, 1050, 1200, 1300, 1450, 1550, 1650
wws_reflectance	list of 8 numbers	WasteWise Scan	Dark-subtracted signal divided by the last white reference
wws_dark_v	number	WasteWise Scan	Mean dark level, a sunlight indicator (see WSC-CAL-001, D)
wws_cal_id , wws_temp_c	number	WasteWise Scan	White reference used, and board temperature
wml_image_ref	text	WasteWise-ml	File name of the consented photo, if kept

On the scanner, each record is stored in a fixed 64-byte binary form (time, sequence, result code, confidence, price grade, flags, pair ID, eight 32-bit reflectance values, dark level, calibration ID, temperature, classifier version and a CRC-16). WSC-CAL-001 section G sizes the log from this. The CSV and JSON Lines exports are generated from it.

■ Decision

None yet. Proposed, awaiting Amish and a joint review with WasteWise-ml. Questions for that review:

1. Does WasteWise-ml accept the shared field names and the pair_id join, or prefer option 2?

2. How is `pair_id` created in the field: typed digits, a sticker, or the phone scanning a code on the scanner screen?
3. Should spectra from consented scans join the WasteWise-ml field dataset, and under which license and ownership terms (WasteWise-ml R14)?

■ Consequences

- If accepted, WasteWise-ml would add the same fields on its side, in its own repo and session.
- R13 stays "at risk" in WSC-CAL-001 until the format is agreed.